How Chinese Foreign Aid Promotes China's Exports

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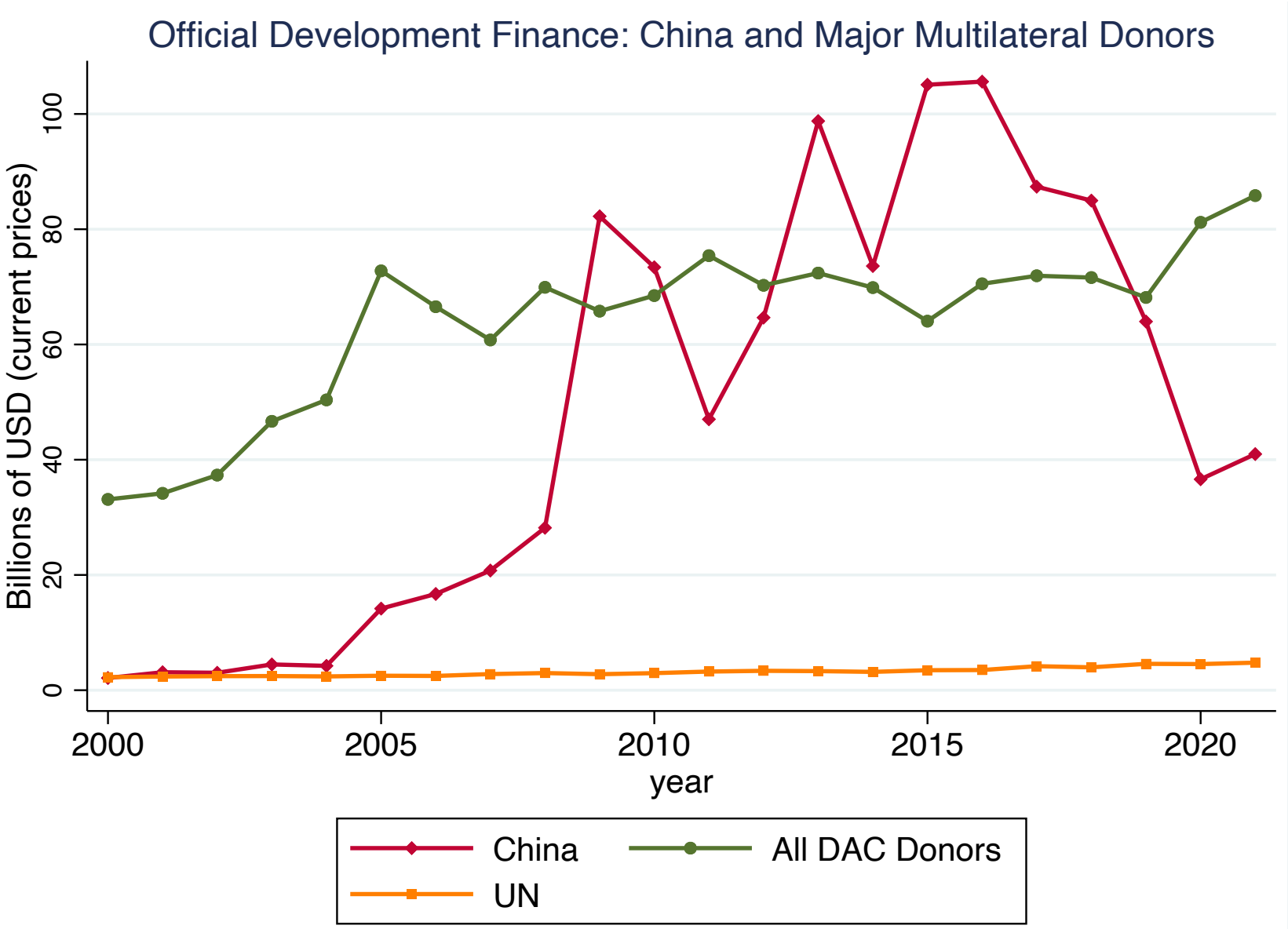
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Motivations

- Over the past two decades, China has witnessed a dramatic increase in its official foreign aid.

- China's official foreign assistance increased from $5.45 billion in 2000 to $40.99 billion in 2021, reflecting an average annual growth rate of 22.1%.

- In recent years, its overseas assistance has even surpassed that of other major donors and lenders (Horn, Reinhart, and Trebesch 2021)


Official Development Finance: China and Major Multilateral Donors
Billions of USD (current prices)
0
20
40
60
80
100
2000
2005
2010
2015
2020
year
China
All DAC Donors
UN

Motivations

- This rapid growth has sparked considerable interest among researchers and policymakers regarding the economic consequences of China's foreign aid.

- Moreover, as a developing country, China offers a unique setting to examine how providing foreign assistance may affect the donor's own development.

- Existing contributions in the literature have primarily focused on examining the effects of Chinese official foreign aid on recipient countries. Much less attention has been paid to its implications for China itself.

- We seek to fill this gap by investigating the domestic economic effects of China's foreign aid. Specifically, we examine how Chinese foreign aid influences China's exports to recipient countries.

What We Do

- We develop a three-country trade framework comprising China, a recipient country, and the rest of the world (ROW). This framework features heterogeneous sectors that differ in their dependence on external finance and their reliance on relationship-specific inputs.

- Aid influences trade patterns through two financial channels. First, aid acts as a form of sovereign endorsement and enhances the creditworthiness of Chinese exporters through government-backed financing arrangements.

- Second, aid generates positive spillovers to the recipient country's domestic financial infrastructure, raising the aggregate enforceability of contracts and relaxing financing constraints for all importers in the recipient country, regardless of their products of origin.

What We Do

- In the presence of financial frictions and incomplete contracts—common features in recipient countries—these aid-induced improvements alleviate both the exporter-side and importer-side wedges in trade transactions.

- The model yields four predictions:
- (i) Chinese aid increases exports from China to the recipient country;
- (ii) this effect is more pronounced in sectors with higher external finance dependence or higher contract intensity;
- (iii) aid also stimulates the recipient country's imports from the ROW;
- and (iv) the spillover to ROW imports is stronger in finance-dependent sectors but invariant to contract intensity.

What We Do

- To test these predictions, we combine detailed project-level data on Chinese official foreign aid with Chinese export data at various levels of granularity.
- We use two complementary strategies to address the endogeneity issue. Our results show that Chinese official aid significantly boosts China's exports to recipient countries and generates positive spillovers by increasing recipient's imports from the rest of the world.
- The positive effect on Chinese exports is stronger in industries with higher dependence on external finance and greater reliance on contract enforcement.
- We rule out several competing explanations, including increased infrastructure-related exports, reductions in transportation costs or information frictions, and increased government purchases.

Contributions to the Literature

- The existing literature primarily examines the impacts of Chinese foreign assistance on recipient countries, exploring effects on:
  - economic growth (Dreher et al., 2021; Bluhm et al., 2025);
  - debt sustainability (Bandiera and Tsiropoulos, 2020);
  - environment (Baehr et al., 2023);
  - politics (Gehring et al., 2022; Isaksson and Kotsadam, 2018).

- **In contrast, our paper focuses on the domestic implications of Chinese official finance from China's perspective.**

Contributions to the Literature

- Similarly, foreign aid literature has assessed impacts on:
  - economic growth (Boone 1996; Burnside and Dollar, 2000; Hansen and Tarp, 2001; Easterly, 2003);
  - local conflicts (Crost et, al., 2014; Nunn and Qian, 2014);
  - politics (Andersen et al., 2022, Birchler et al., 2016);
  - firm growth (Chauvet and Ehrhart, 2018).
- However, much of the existing literature focuses on aid from international organizations (e.g., the World Bank and IMF) or developed-country donors (e.g., the US). **Our study investigates foreign aid from a developing country-China.**

Outline

- Theoretical Model
- Data and Empirical Strategy
- Empirical Results
  - Baseline Results
  - Robustness Checks
  - Evidence from Disaggregated (Industry/Product-Level) Data
- Conclusions

Theoretical Model: Setup

- Three countries: China (C), recipient country (R), the rest of the world (W)

- Each country consists of k sectors. In each sector, the final good aggregates heterogeneous varieties through a CES aggregator: $Q_k = \left(\int_{\omega \in \Omega_k} q\left(\omega\right)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}$

- Country R can source varieties $q(\omega)$ either domestically or through imports from China or the ROW

- The market operates under monopolistic competition. Sectors differ in:
  - External finance dependence $d_k \in (0,1)$
  - Relationship specificity $\rho_k \in (0,1)$

Theoretical Model: Setup

The supply-side structure follows the framework of Krugman (1980). In each country $i \in \{C, R, W\}$, there are N_i firms with homogenous productivity that sell to global markets.

Free entry determines N_i endogenously through firms' profits across all markets. For the subsequent analysis, we focus on a partial equilibrium within a single sector k in country R, suppressing the sector index for notational simplicity.

General equilibrium variables—including wages w_i, the mass of active firms N_i, and real aggregate expenditure $E_R = Y_R P_R^{\sigma-1}$—are treated as exogenous.

Theoretical Model: Firm Problem

A producer in country $i \in \{C, R, W\}$ manufactures a single differentiated variety (suppressing the ω index) with quantity q_{iR} destined for the market in country R.

Production follows a Leontief technology: $q_{iR} = \min\{m_{iR}^{RS}/\rho, m_{iR}^{NS}/(1-\rho)\}$, where m_{iR}^{RS} denotes relationship-specific inputs and m_{iR}^{NS} denotes non-relationship-specific (generic) inputs. The parameter $\rho \in (0,1)$ represents the input share for relationship-specific components and serves as a standard proxy for contract intensity.

Each producer is randomly paired with one domestic input supplier in country i and one foreign retailer in country R. Both suppliers and retailers operate in perfectly competitive sectors. The three parties engage in a sequential contracting process, structured as follows.

Theoretical Model: Timeline

- The producer's optimization problem incorporates both financial frictions and contractual incompleteness. The timeline unfolds as follows:

(1) The producer makes a take-it-or-leave-it contract offer to the supplier, specifying an upfront transfer t_{iR} (which may be positive or negative). Due to contractual incompleteness, input quantities cannot be verified or contracted upon ex ante.

(2) The supplier accepts the offer (along the equilibrium path) and subsequently chooses input quantities (Leontief technology: $m_{iR}^{RS}/m_{iR}^{NS} = \rho/(1-\rho)$). The unit cost for both input types is c_i.

Theoretical Model: Timeline

(3) The producer and supplier bargain over the expected export revenue x_{iR}. Because relationship-specific inputs have zero salvage value outside the partnership, only the non-relationship-specific inputs can be redeployed. We assign property rights (the right of disposal) over these generic inputs to the producer. Consequently, the producer's outside option is $(1-\rho)c_i q_{iR}$, while the supplier's outside option is zero. The surplus (quasi-rent) available for division is therefore $s_{iR} = x_{iR} - (1-\rho)c_i q_{iR}$.

(4) Retailer pays producer x_{iR}, and producer and supplier split it.

(5) The producer ships the goods to the retailer who sells them to the consumers. Due to iceberg trade costs $\tau_{iR} > 1$, only q_{iR}/τ_{iR} units arrive in country R. The demand function is given by $q_{iR}/\tau_{iR} = E_R p_{iR}^{-\sigma}$ where p_{iR} denotes the consumer price. The retailer's total sales revenue is then $r_{iR} = p_{iR} q_{iR}/\tau_{iR} = E_R^{\frac{1}{\sigma}} \tau_{iR}^{\frac{1-\sigma}{\sigma}} q_{iR}^{\frac{\sigma-1}{\sigma}}$

Theoretical Model: Two-Side Financial Frictions

- Importer-side friction
  - Retailer needs raise *d* fraction of the retailer payment x_{iR} externally, and pledge a repayment l_{iR} to lenders.
  - Lenders in country *R* enforce only $\kappa_R \in (0,1)$ of the promised payments, reflecting the institutional quality of the destination market.
  - Normalizing the risk-free rate to zero, the retailer must pledge a repayment amount l_{iR} such that the lender's expected recovery $\kappa_R l_{iR}$ covers dx_{iR}. Thus, the participation constraint requires $\kappa_R l_{iR} \geq dx_{iR}$, yielding an equilibrium pledge of $l_{iR} = dx_{iR}/\kappa_R$. The retailer's total outlay for the export shipment therefore becomes $(1-d)x_{iR} + dx_{iR}/\kappa_R = (1-d+d/\kappa_R)x_{iR}$.
  - The zero-profit condition implies: $x_{iR} = \frac{1}{1-d+\frac{d}{\kappa_R}} r_{iR} = \Lambda_R^I r_{iR}$, where $\Lambda_R^I \in (0,1)$ represents the importer-side financial wedge.

Theoretical Model: Two-Side Financial Frictions

- Exporter-side friction
  - Producer also needs to raise *d* fraction of production cost $c_i q_{iR}$ externally
  - The pledgeability rate $\theta_{iR} \in (0,1)$, captures the lender's assessment of the creditworthiness and institutional quality of export destination
  - Following a similar derivation, the total cost incurred in production equals $(1 - d + d/\theta_{iR}) c_i q_{iR} = c_i q_{iR} / \Lambda_{iR}^{E}$, where $\Lambda_{iR}^{E} \in (0,1)$ is exporter-side financial wedge
  - Note that this wedge operates as a sunk cost by the time the producer and supplier engage in ex-post Nash bargaining. Because external financing and input procurement strictly precede the bargaining stage, the incurred financing cost $c_i q_{iR}(1/\Lambda_{iR}^{E} - 1)$ cannot be recovered or renegotiated.

Theoretical Model: Equilibrium

- We solve the equilibrium via backward induction. The retailer collects gross sales revenue r_{iR} and pays the trade value $x_{iR} = \Lambda_R^I r_{iR}$ to the producer.

- We assume that the relative bargaining power of the producer is $\beta \in (0,1)$. So, the producer gets $\beta s_{iR} + (1-\rho)c_i q_{iR}$, and the supplier gets $(1-\beta)s_{iR} = (1-\beta)x_{iR} - (1-\beta)(1-\rho)c_i q_{iR}$.

- Supplier's problem: $\underset{q_{iR}}{max}\ \pi_{iR}^S = (1-\beta)x_{iR} - (1-\beta)(1-\rho)c_i q_{iR} - \frac{1}{\Lambda_{iR}^E} c_i q_{iR}$
- The FOC yields: $q_{iR} = \frac{1}{c_i}\frac{\sigma-1}{\sigma}\Lambda_R^I \frac{1-\beta}{\frac{1}{\Lambda_{iR}^E}+(1-\beta)(1-\rho)} r_{iR} = \frac{1}{c_i}\frac{\sigma-1}{\sigma}\Lambda_R^I \tilde{\Lambda}_{iR}^E r_{iR}$

Theoretical Model: Equilibrium

- Demand function: $q_{iR} = \frac{1}{c_i}\frac{\sigma-1}{\sigma}\Lambda_R^I \frac{1-\beta}{\frac{1}{\Lambda_{iR}^E}+(1-\beta)(1-\rho)} r_{iR} = \frac{1}{c_i}\frac{\sigma-1}{\sigma}\Lambda_R^I \tilde{\Lambda}_{iR}^E r_{iR}$
- Sales revenue: $r_{iR} = \left(\frac{\sigma-1}{\sigma}\right)^{\sigma-1} E_R \tau_{iR}^{1-\sigma} c_i^{1-\sigma} (\Lambda_R^I)^{\sigma-1} (\tilde{\Lambda}_{iR}^E)^{\sigma-1}$
- Sectoral aggregate exports: $X_{iR} = N_i x_{iR} = N_i \Lambda_R^I r_{iR}$

Theoretical Model: The Effects of Aid on Imports from China

- Suppose $\theta_{CR} = \theta(A_{CR})$, $\kappa_R = \kappa(A_{CR})$, both increase with Chinese aid to country *R*
- θ_{RR} and θ_{WR} remain constant in country *R* and ROW, respectively
- When $\Delta A_{CR} > 0$, the change in China-to-R exports is:

$$\Delta \ln X_{CR} = (\sigma - 1)\Delta \ln \tilde{\Lambda}^{E}_{CR} + \sigma \Delta \ln \Lambda^{I}_{R} = (\sigma - 1)\frac{\partial \ln \tilde{\Lambda}^{E}_{CR}}{\partial \theta_{CR}}\Delta \theta_{CR} + \sigma \frac{\partial \ln \Lambda^{I}_{R}}{\partial \kappa_R}\Delta \kappa_R > 0$$

Proposition 1. An increase in Chinese aid to country R increase China-to-R exports.

Theoretical Model: The Effects of Aid on Imports from China

- Chinese export value and external finance dependence:

$$\frac{\partial \Delta \ln X_{CR}}{\partial d} = (\sigma - 1)\frac{\partial^2 \ln \tilde{\Lambda}_{CR}^{E}}{\partial \theta_{CR} \partial d}\Delta\theta_{CR} + \sigma \frac{\partial^2 \ln \Lambda_{R}^{I}}{\partial \kappa_R \partial d}\Delta\kappa_R > 0$$

- Chinese export value and contract intensity:

$$\frac{\partial \Delta \ln X_{CR}}{\partial \rho} = (\sigma - 1)\frac{\partial^2 \ln \tilde{\Lambda}_{CR}^{E}}{\partial \theta_{CR} \partial \rho}\Delta\theta_{CR} + \sigma \frac{\partial^2 \ln \Lambda_{R}^{I}}{\partial \kappa_R \partial \rho}\Delta\kappa_R > 0$$

Proposition 2. The positive effect of Chinese aid on China-to-Rexports is stronger for sectors with higher external financial dependence and contract intensity.

Theoretical Model: The Effects of Aid on Imports from ROW

- ROW's exports to country R is: $\Delta \ln X_{WR} = \sigma \Delta \ln \Lambda_R^I = \sigma \frac{\partial \ln \Lambda_R^I}{\partial \kappa_R} \Delta \kappa_R$

Proposition 3. ***An increase in Chinese aid to country R raises rest-of-world exports to country R.***

Theoretical Model: The Effects on Imports from ROW

- ROW's exports and external finance dependence:

$$\frac{\partial \Delta \ln X_{WR}}{\partial d} = \sigma \frac{\partial^2 \ln \Lambda_R^I}{\partial \kappa_R \partial d} \Delta \kappa_R > 0$$

- ROW's exports and contract intensity:

$$\frac{\partial \Delta \ln X_{WR}}{\partial \rho} = \sigma \frac{\partial^2 \ln \Lambda_R^I}{\partial \kappa_R \partial \rho} \Delta \kappa_R = 0$$

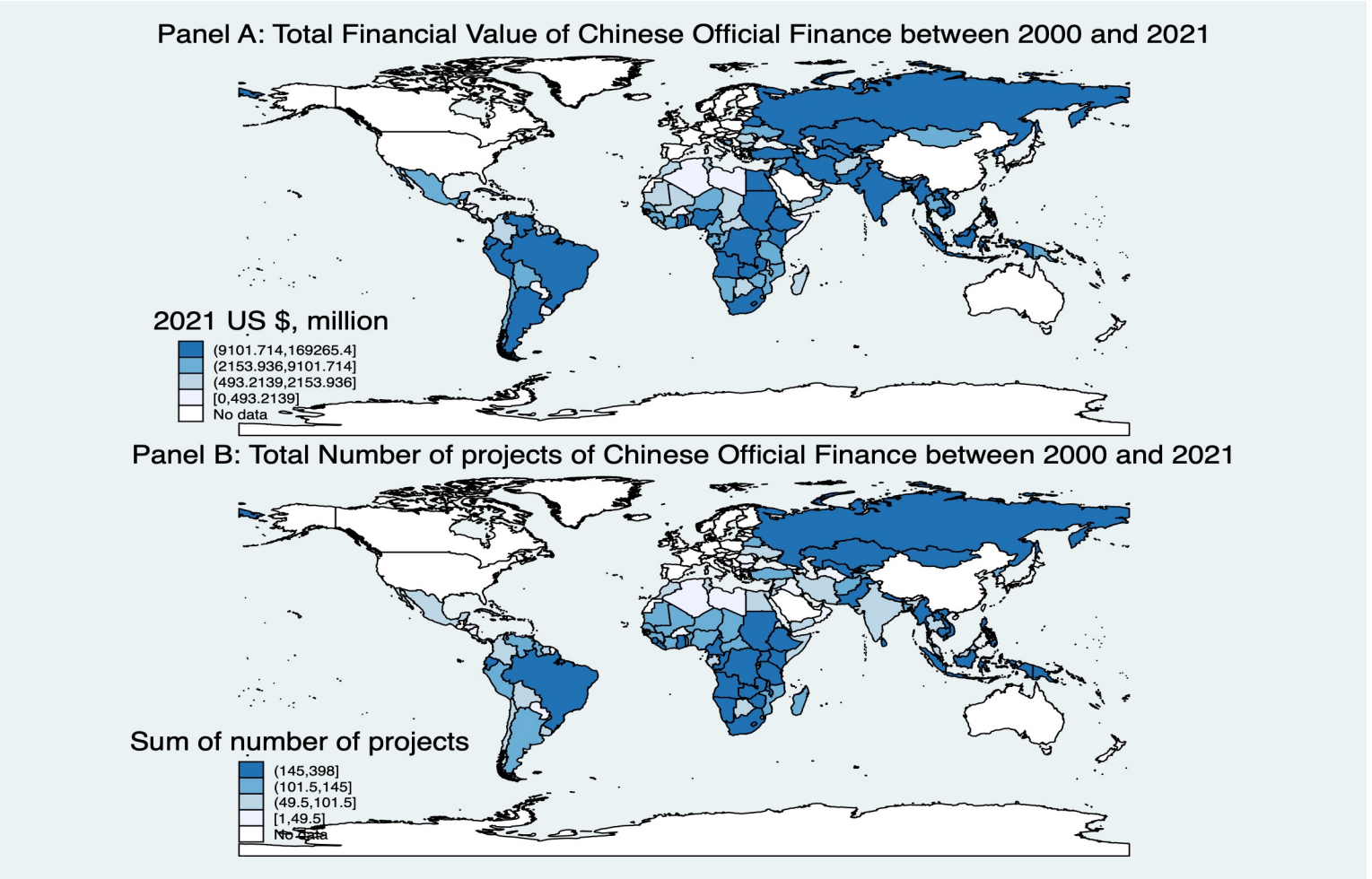
Proposition 4. The effect of Chinese aid on country R's imports from the ROW is larger in sectors with greater external finance dependence, but invariant to relationship specificity.

Data

- Chinese official aid data: AidData 3.0 covering all Chinese government-financed projects worldwide from 2000 to 2021. It includes recipient country, commitment year, financial value, and project description.
- Exclude projects identified only at the regional level (e.g., Africa or Asia) rather than for a specific country, as well as multi-country projects. We also remove projects that were double-counted or subsequently canceled.
- Also exclude COVID-19–related projects, including those involving information dissemination, testing, prevention, immunization, or medical supplies.
- After applying these restrictions, our final sample consists of 15,906 projects.

Data

Figure1. Geographical Distribution of Chinese Official Aid



Data

- Dependent variable: China's annual export value to recipient country, at both country-year or at country-product/industry-year levels (source: UN comtrade).
- Control variables: recipient country's real GDP, real GDP growth rate, population, direct exchange rate, and tariff rate (source: UN, UN comtrade).

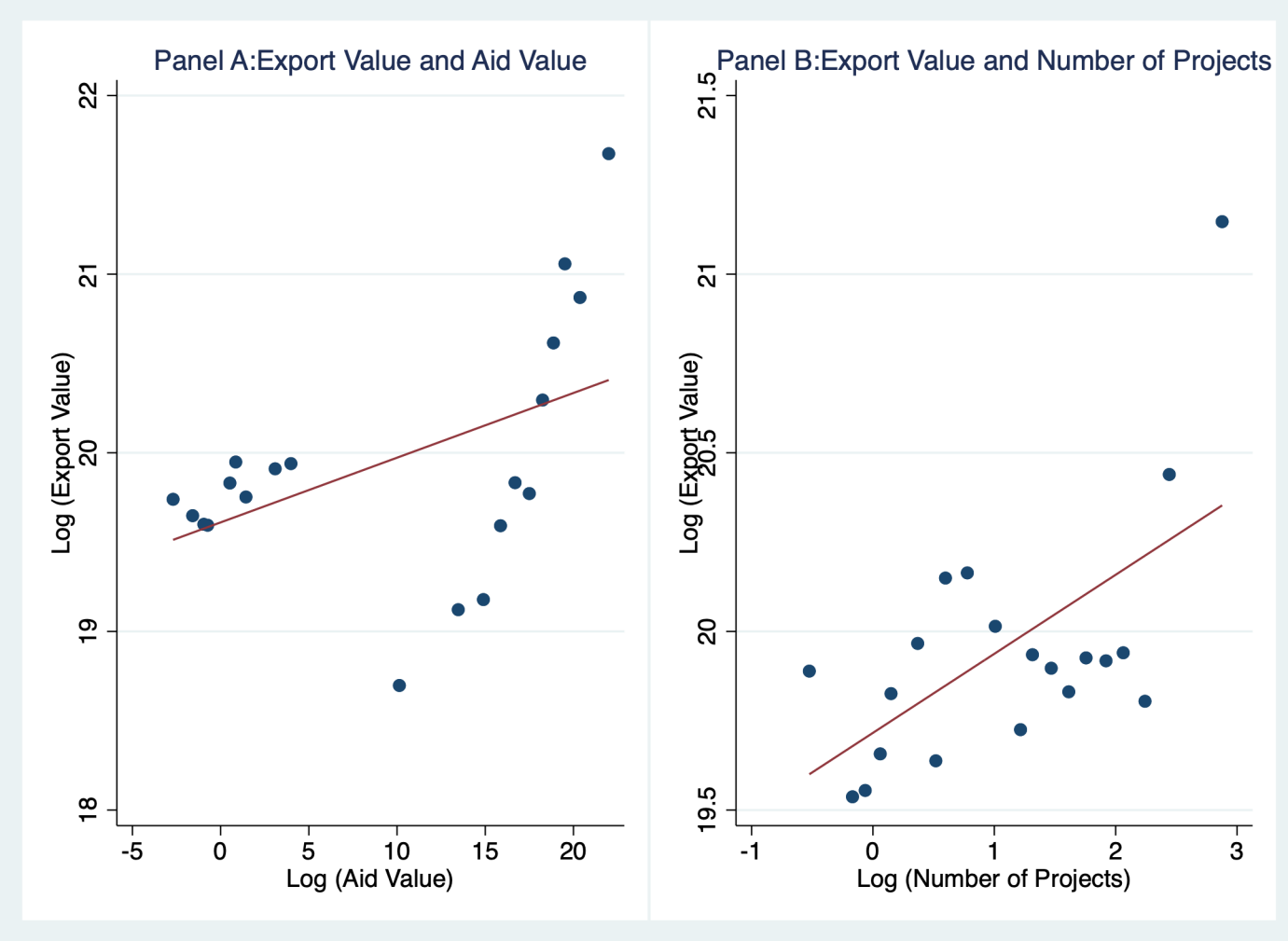
Empirical Specification

$$y_{ct} = \beta_0 + \beta_1 OF_{ct} + \beta_x X_{ct-1} + \varphi_t + \varphi_c + \varepsilon_{ct}$$

- y_{ct}: log of China's export value to recipient country c in year
- OF_{ct}: log of Chinese official aid commitment
- X_{ct-1}: set of country-level control variables, including real GDP (log), real GDP growth rate, population (log), direct exchange rate, and average tariff rate (log)
- φ_t: year fixed effect, capturing common time-varying shocks across countries
- φ_c: country fixed effect, accounting for time-invariant country-specific factors

Empirical Specification

Figure2. Chinese Foreign Aid and Exports to Recipient



Endogeneity Issue: IV Approach

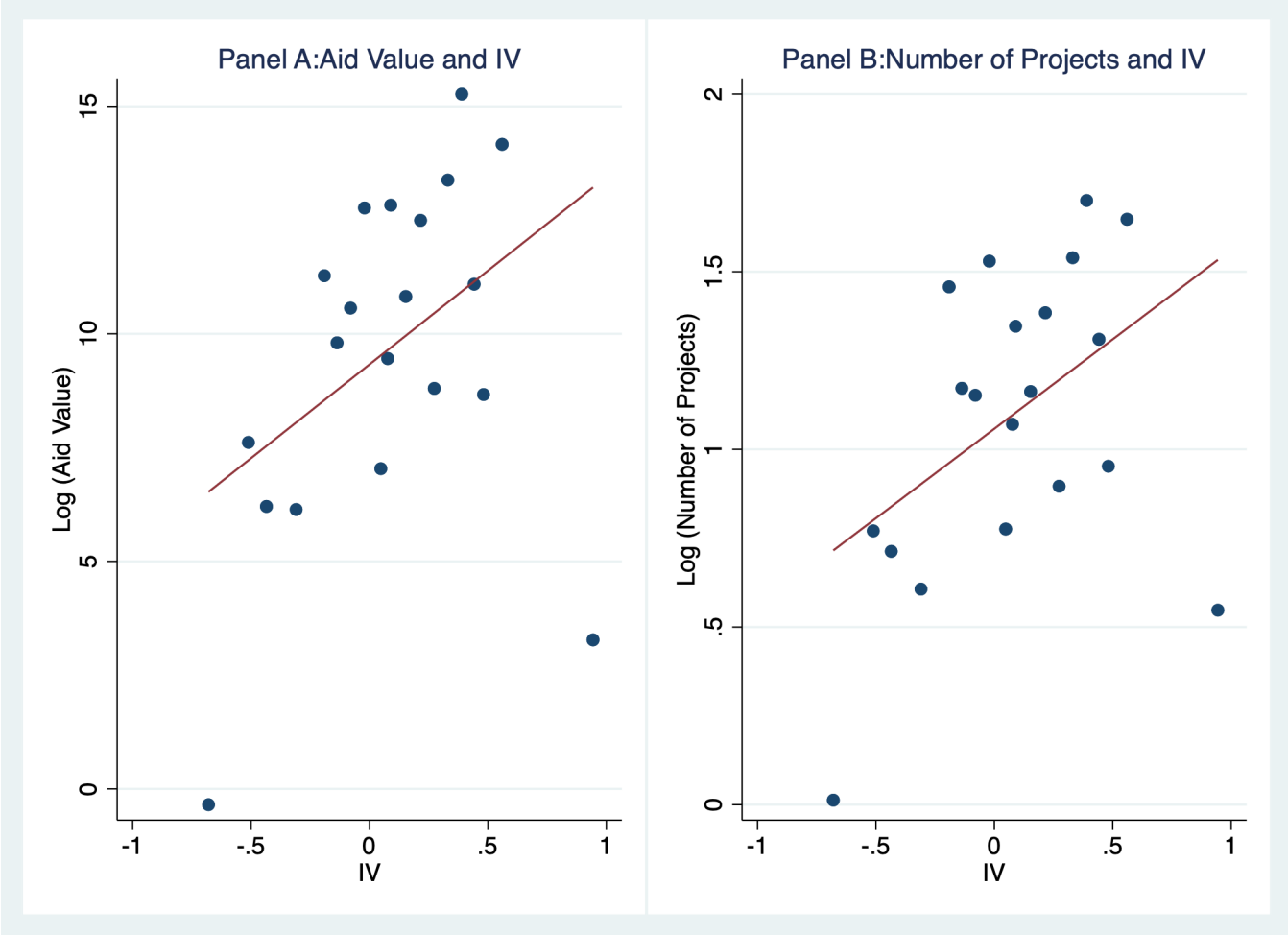
IV method (following Dreher et al., 2021, and Nunn and Qian, 2014):

$$IV_{ct} = P_c \times X_{t-1}$$

- P_c: probability of receiving Chinese official aid, based on aid receiving probability in 2000-2021, 2000-2010, and 1975-1989
- X_{t-1}: supply-side shock to Chinese aid capacity, using the PCA method to extract an input factor index of the production of six input materials (iron, steel, timber, aluminum, cement and glass)

Endogeneity Issue: Relevance of IV

Figure3. Chinese Foreign Official Aid and the IV



The Effect of Chinese Aid on China's Exports

Table 1. Baseline IV Regression Results

	(1) Baseline IV	(2) Alternative IV1	(3) Alternative IV2
Panel A: Second-stage results			
Log aid value	0.095***	0.094**	0.088*
	(3.122)	(2.469)	(1.730)
Log GDP	0.373	0.371	0.365
	(0.975)	(0.973)	(0.916)
GDP_growth	0.007	0.007	0.007
	(1.105)	(1.057)	(1.025)
Log Population size	0.781	0.786	0.807
	(1.004)	(1.014)	(0.986)
Exchange rate	0.040	0.037	0.025
	(0.220)	(0.195)	(0.136)
Log Tariff rate	-0.147	-0.146	-0.143
	(-1.129)	(-1.126)	(-1.108)
Panel B: First-stage results			
IV	1.830***	1.596***	1.719***
	(6.642)	(5.148)	(2.934)
No. of Obs.	2,153	2,153	2,153
Kleibergen-Paap F Statistic	44.121	26.506	8.606
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Robustness

- Using an alternative measure of Chinese official aid
- Excluding the aid categories as buyer's credit or supplier's credit
- Controlling aid from other donors (DAC, World Bank, UN)
- Dropping outliers (top and bottom 1%)
- **Addressing concerns related to the exclusion restriction of IV:**
  - Direct effects: we drop exports from industries that use the six materials (steel, iron, cement, aluminum, timber, glass) as major inputs
  - Indirect effects: we further identify downstream industries that use the above industries as main inputs in their production and drop their exports as well
  - Controlling for Chinese GDP growth rate×country fixed effects

Robustness

Table 2. Robustness Checks

	(1) Alternative measure of aid	(2) Exclude buyer or supplier credit	(3) Control aid from other donors	(4) Drop top and bottom 1%	(5) Exclude indirect effect	(6) Exclude 2nd order indirect effect	(7) Control growth ×country FE
Log no. of projects	0.851***						
	(3.109)						
Log aid value		0.096***	0.065**	0.085**	0.094***	0.085***	0.076***
		(3.078)	(2.385)	(2.497)	(3.062)	(2.844)	(2.938)
Log DAC value			0.022***				
			(3.419)				
Log IBRD-IDA value			-0.026*				
			(-1.948)				
Log UN value			-0.004				
			(-0.444)				
No. of Obs.	2,153	2,153	2,153	1,968	2,153	2,152	2,153
Kleibergen-Paap F Statistic	45.663	43.822	41.051	40.152	44.121	43.882	47.665
Country-year Level Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	No
China GDP growth×country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Recipients' Imports from the Rest of the World

Table 3. Trade Creation or Trade Diversion

	(1) Total imports from the world	(2) Imports from the ROW	(3) Imports from China/total impo
Log aid value	0.021**	0.017*	0.004*
	(1.995)	(1.731)	(1.974)
No. of Obs.	2,153	2,153	2,153
Kleibergen-Paap F Statistic	44.121	44.121	44.121
Country-year Level Controls	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Endogeneity Issue: Direct Evidence on the Mechanisms

- Our second identification strategy follows the approach of Rajan and Zingales (1998), who emphasize that direct evidence on the underlying mechanisms can serve as a “smoking gun” for causality.
- Using more disaggregated country–industry data, we test whether Chinese foreign aid disproportionately promotes exports in sectors that rely more heavily on external finance or contract enforcement, as predicted by our model.
- We also examine whether aid disproportionately increases recipient imports from the ROW in finance-dependent sectors.
- In addition, we offer micro-level evidence to rule out alternative explanations for the observed export expansion. The direct evidence on the mechanisms can thus offer more confidence in causality.

Evidence from Disaggregated Data and Mechanisms

To examine the underlying mechanisms, we exploit cross-industry variation in reliance on external finance and contract enforcement. Specifically, we estimate the following country-industry-level regression:

$$\ln(Export_{ict}) = \beta_0 + \beta_1\, OF_{ct} \times financedep_i + \beta_2 OF_{ct} \times contract_i$$

$$+\mu X_{ct-1} + \rho tariff_{cit-1} + \alpha_{ic} + \gamma_{ct} + \varepsilon_{ict}$$

- $financedep_i$ is a dummy variable equal to one if an industry's external finance dependence exceeds the sample mean
- $contract_i$ is a dummy variable equal to one if an industry's contract intensity exceeds the sample mean
- α_{ic} is the country-industry or product fixed effect
- γ_{ct} is the country-year fixed effect

Evidence from Disaggregated Data and Mechanisms

Table 4: Country-Industry Level Evidence on Mechanisms

	Imports from China				Imports from the ROW			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A								
Log aid value	0.075***	0.070***			0.017*	0.021*		
	(3.345)	(2.756)			(1.779)	(1.844)		
Log aid value×extfin			0.039***	0.027*			0.012***	0.015***
			(3.448)	(1.975)			(2.997)	(2.693)
Log aid value×contract			0.075***	0.058***			0.004	-0.004
			(4.668)	(3.211)			(0.625)	(-0.537)
Kleibergen-Paap F Statistic	47.733	26.516	21.426	11.418	47.731	26.512	21.426	11.418
Panel B								
Log no. of projects	0.694***	0.789**			0.157*	0.236*		
	(3.240)	(2.393)			(1.810)	(1.806)		
Log no. of projects×extfin			0.402***	0.375*			0.125***	0.205**
			(3.522)	(1.879)			(2.939)	(2.387)
Log no. of projects×contract			0.784***	0.797***			0.044	-0.055
			(4.528)	(2.752)			(0.667)	(-0.490)
Kleibergen-Paap F Statistic	41.232	15.566	15.880	5.026	41.233	15.566	15.880	5.026
No. of Obs.	71,186	71,186	71,178	71,178	71,185	71,185	71,177	71,177
Country-industry time trend	No	Yes	No	Yes	No	Yes	No	Yes
Country-level Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No	Yes	Yes	No	No
Country×Ind FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country×Year FE	No	No	Yes	Yes	No	No	Yes	Yes

Infrastructure-related Products

- A first alternative explanation is that Chinese aid projects, which are often linked to infrastructure and construction, may favor procurement from Chinese firms, suggesting that the observed increase in Chinese exports may be solely driven by infrastructure-related goods.
- First, using China's 2007 Input-Output table, we identify the five sectors that supply the largest share of inputs to the construction sector and drop all products mapped to these sectors. If infrastructure imports are driving the results, the estimated aid effect should disappear once such products are excluded.
- Second, following Chor et al. (2021), we classify products as upstream if their index exceeds the sample mean and downstream if otherwise. If procurement for aid projects explains our results, the aid effect should be stronger for upstream products.

Infrastructure-related Products

Table 5. Evidence on the Infrastructure Inputs Explanation

	Country-level	Country-industry-level	Country-product-level	Upstream products	Downstream products
	(1)	(2)	(3)	(4)	(5)
Panel A					
Log aid value	0.093***	0.068***	0.059***	0.048**	0.071***
	(3.028)	(3.073)	(2.836)	(2.376)	(3.287)
Kleibergen-Paap F Statistic	44.121	46.920	26.964	25.535	29.229
Panel B					
Log no. of projects	0.827***	0.635***	0.687**	0.567**	0.830***
	(3.023)	(2.989)	(2.464)	(2.129)	(2.793)
Kleibergen-Paap F Statistic	45.663	40.057	10.876	10.173	12.216
No. of Obs.	2,153	70,166	3,136,100	1,840,947	1,549,214
Country-level Controls	Yes	Yes	Yes	Yes	Yes
Country×Product FE	No	No	Yes	Yes	Yes
Country×Industry FE	No	Yes	No	No	No
Country FE	Yes	No	No	No	No
Year FE	Yes	Yes	Yes	Yes	Yes

Reduction of Transportation Costs

- A second potential explanation is that the rise in Chinese exports is driven by reduced transportation costs, given that Chinese official aid is often linked to infrastructure projects.
- We use country-product level data and construct two product-level proxies for sensitivity of transportation costs. The first one is the value-to-weight ratio, where higher values imply lower sensitivity to transportation cost. The second one is cif-to-fob ratio (Anderson and van Wincoop, 2004), where higher value indicate greater transport cost sensitivity.
- We interact each measure with Chinese official aid. If reduced transportation cost is the primary channel, the aid effect should be stronger for products that are more sensitive to transportation costs.

Reduction of Transportation Costs

Table 6. Evidence on the Reduction of Transportation Cost Mechanism

	Value-to-weight ratio		Cif-to-fob ratio	
	(1)	(2)	(3)	(4)
Log aid value	0.054***		0.058***	
	(2.725)		(2.859)	
Log aid value×ratio	0.008	0.006	0.068	0.072
	(0.664)	(0.458)	(0.537)	(0.588)
No. of Obs.	2,131,270	2,131,265	3,396,605	3,396,598
Kleibergen-Paap F Statistic	12.978	728.104	13.676	37.850
Country-year Level Controls	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No
Country×Product FE	Yes	Yes	Yes	Yes
Country×Year FE	No	Yes	No	Yes

Reduction of Information Frictions

- The third possible explanation is that Chinese official aid reduces information frictions, aiding potential Chinese exporters in gaining knowledge about the recipient countries' markets.

- First, using country-level data, we interact Chinese official aid with proxies for information barriers (geographic distance and linguistic proximity) to examine whether aid effects are stronger in countries with higher information frictions.

- Second, using country-product level data, we construct a normalized revealed comparative advantage (NRCA) index following Yu et al. (2009) and test whether aid disproportionately promotes Chinese exports in products where China has a comparative advantage.

Reduction of Information Frictions

Table 7. Evidence on the Reduction of Information Frictions Mechanism

	Country-level evidence		Country-product-level evidence	
	(1)	(2)	(3)	(4)
Log aid value	-0.104	0.086*	0.110***	
	(-0.350)	(1.889)	(3.927)	
Log aid value×Distance	0.022			
	(0.705)			
Log aid value×Linguistic proximity		-0.040		
		(-0.463)		
Log aid value×RCA			-0.068***	-0.073***
			(-4.432)	(-4.540)
RCA			1.011***	1.068***
			(6.461)	(6.361)
No. of Obs.	2,101	1,974	1,898,514	1,898,479
Kleibergen-Paap F Statistic	17.510	17.262	10.588	24.296
Country-year Level Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No
Country FE	Yes	Yes	No	No
Country×Product FE	No	No	Yes	Yes
Country×Year FE	No	No	No	Yes

Government Purchase

- The last possible explanation is that Chinese aid enhances China's influence over the government procurement decisions of recipient countries.

- First, at the country level, we include an interaction term between Chinese official aid and the recipient country's government expenditure-to-GDP ratio to test whether aid effects are stronger in countries with higher government spending.

- Second, at the country-industry level, we utilize SITC 2-digit industry-level measures of government purchases and imports (constructed by Berger et al., 2013) to examine whether the effect of aid is more pronounced in industries with higher levels of government procurement.

Government Purchase

Table 8. Evidence on the Government Purchase Mechanism

	Country-level data	Country-industry level data			
	(1)	(2)	(3)	(4)	(5)
Log aid value	0.082	0.069***		0.072***	
	(1.252)	(3.038)		(3.144)	
Log aid value×gov_share	0.000				
	(0.049)				
Log aid value×high_purchase		-0.005	-0.007		
		(-0.450)	(-0.630)		
Log aid value×high_import				-0.017	-0.023**
				(-1.625)	(-2.120)
No. of Obs.	1,915	104,743	104,734	104,743	104,734
Kleibergen-Paap F Statistic	7.465	22.770	43.758	22.799	45.926
Country-year Level Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	Yes	No
Country FE	Yes	No	No	No	No
Country×Ind FE	No	Yes	Yes	Yes	Yes
Country×Year FE	No	No	Yes	No	Yes

Conclusions

- Chinese official aid significantly promotes China's exports to recipient countries, consistent with our theoretical prediction.
- We also document positive spillover effects: Chinese aid increases recipient's countries' imports from the rest of the world.
- The export-promoting effect is stronger in industries with higher external finance dependence and more reliance on contract enforcement.